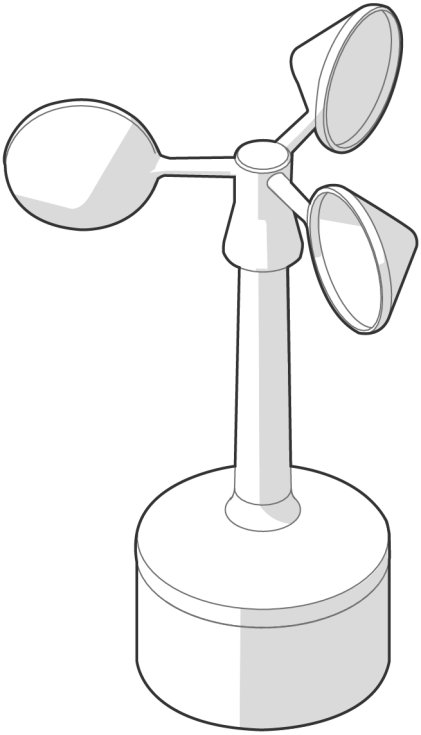
M210293EN-D

User Guide

Vaisala Anemometer

WAA151



VAISALA

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1. About this document

1.1 Version information

Table 1 Document versions (English)

Document code	Date	Description
M210293EN-D	August 2023	Updated the distance constant value. Corrected the spare part codes (1433WA).
M210293EN-C	November 2018	Updated spare part descriptions of ZZ45037 and ZZ45037SPEC.
M210293EN-B	August 2018	Installation instructions updated. Connector type and item code changed.

1.2 Related manuals

Table 2 Related manuals

Document code	Name
M210349EN	WAC151 User Guide (M210349EN)
M210294EN	WAV151 User Guide (M210294EN)
M210822EN	WAC155 User Guide (M210822EN)

1.3 Documentation conventions

WARNING! Warning alerts you to a serious hazard. If you do not read and follow instructions carefully at this point, there is a risk of injury or even death.

CAUTION! Caution warns you of a potential hazard. If you do not read and follow instructions carefully at this point, the product could be damaged or important data could be lost.

Note highlights important information on using the product.



Tip gives information for using the product more efficiently.



Lists tools needed to perform the task.



Indicates that you need to take some notes during the task.

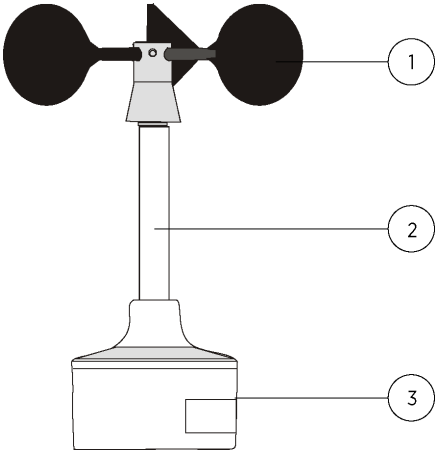
1.4 Trademarks

Vaisala® is a registered trademark of Vaisala Oyj.

All other product or company names that may be mentioned in this publication are trade names, trademarks, or registered trademarks of their respective owners.

2. Product overview

2.1 Vaisala Anemometer WAA151



1 Cup wheel assembly
2 Sensor shaft
3 Lower body

WAA151 is a fast-response, low-threshold anemometer. Three lightweight, conical cups mounted on the cup wheel provide a linear response rate over the entire operating range, up to 75 m/s (168 mph).

A wind-rotated chopper disc attached to the shaft of the cup generates a pulse output from a phototransistor in the base of the sensor. The output pulse rate is directly proportional to wind speed.

The heating element in the shaft tunnel keeps the temperature of the bearings above the freezing level in cold climates. The nominal heating power is 10 W. Vaisala recommends that you use a thermostat switch in the cross-arm for switching the heating power on below +4 °C (+39 °F).

2.2 Safety

The WAA151 anemometer delivered to you has been tested for safety and approved as shipped from the factory. Note the following precautions:

WARNING! Ground the product and verify outdoor installation grounding periodically. Failure to provide proper grounding can result in injury or death from electrical shock and can severely damage the equipment.

CAUTION! Improper modification can damage the product or lead to malfunction. Any modification voids your warranty.

2.2.1 ESD protection

Electrostatic discharge (ESD) can damage electronic circuits. Vaisala products are adequately protected against ESD for their intended use. However, it is possible to damage the product by delivering electrostatic discharges when touching, removing, or inserting any objects in the equipment housing.

To avoid delivering high static voltages to the product:

- Handle ESD-sensitive components on a properly grounded and protected ESD workbench or by grounding yourself to the equipment chassis with a wrist strap and a resistive connection cord.
- If you are unable to take either precaution, touch a conductive part of the equipment chassis with your other hand before touching ESD-sensitive components.
- Hold component boards by the edges and avoid touching component contacts.

3. Installation

3.1 Selecting location

Finding a suitable site for the product is important for getting representative ambient measurements. Make sure that the site represents the general area of interest.

Allow sufficient clearance for the wind sensors. Do not install wind sensors next to a building or any other object that can affect the flow of air.

CAUTION! Installations on top of high buildings or masts and in sites on open grounds are vulnerable to lightning strikes. A nearby lightning strike may induce a high-voltage surge not tolerable by the internal surge suppressors of the instrument. Additional protection is needed in regions with frequent, severe thunderstorms, especially when long line cables (>30 m (98 ft)) are used. Vaisala recommends using a surge protector, such as WSP150, in all sites with an elevated risk of a lightning strike.

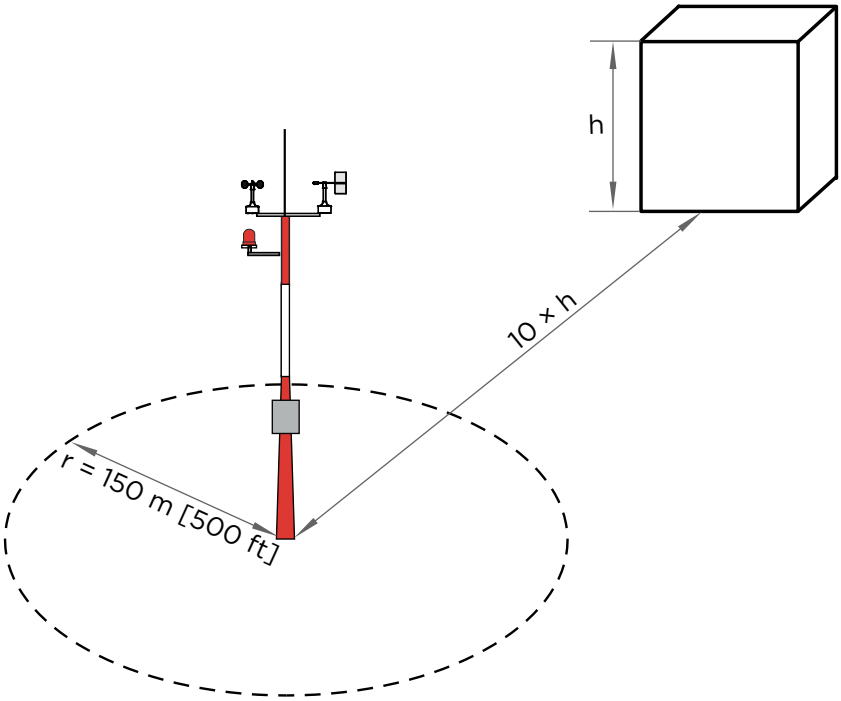


Figure 1 Recommended mast location in open area

Any object of height (h) does not remarkably disturb wind measurement at a minimum distance of 10 × h. There must be at least 150 m (500 ft) open area in all directions from the mast. Minimum distance between the mast and obstacles is ten times the height of an obstacle.

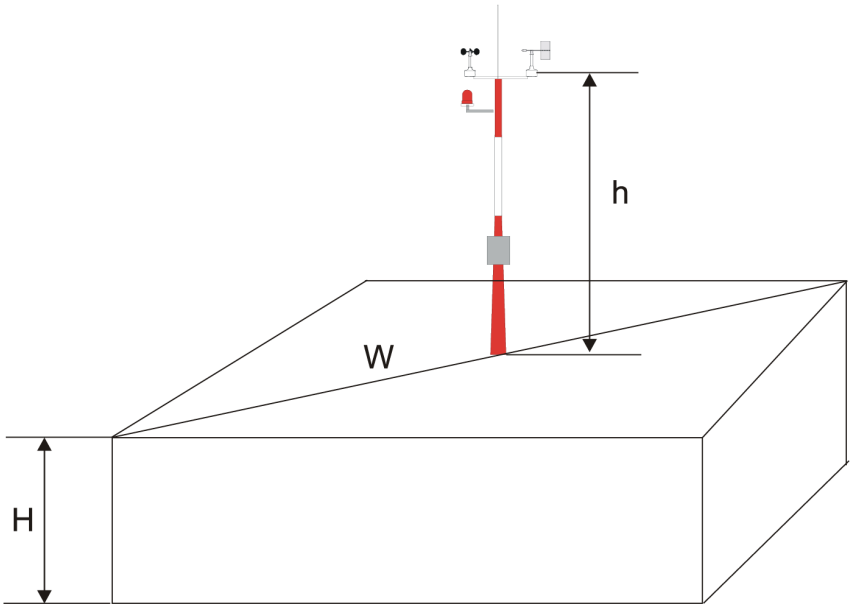


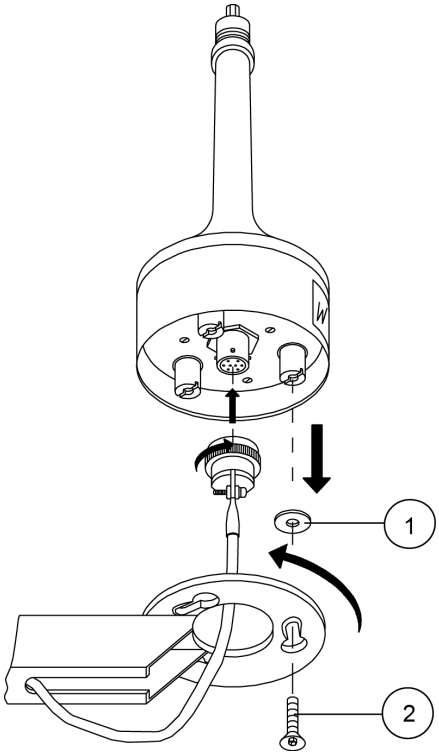
Figure 2 Recommended mast length on top of building

The recommended minimum length (h) for the mast that is installed on the top of a building is 1.5 times the height of the building (H). When the diagonal (W) is less than the height (H) the minimum length of the mast is 1.5 × W.

3.2 Mounting wind sensors on cross-arm

- 2-mm Allen key
- Compass

1. Fit the cable plug through the mounting flange at the end of the cross-arm, and connect the cable to the sensor.



1 Plastic washer
2 Hex screw

2. Insert the plastic washers between the flange and the sensor.
3. Face the sensor product label South, if possible, and mount the sensor to the flange by twisting.
4. Tighten the screws.
5. Mount the assembly and tighten the fixing screw.

3.3 Connecting the anemometer

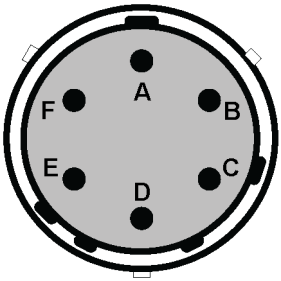


Figure 3 WAA151 connector sensor side

A F+, power input from 9.5 to 15.5 V DC
B GND, common ground
C Fout, signal output
D HTNG, 20 V DC or V AC
E HTNG, 20 V DC or V AC
F Not connected

Vaisala recommends using the SOURIAU UTS6JC10E6P cable connector for the cable side of this sensor.

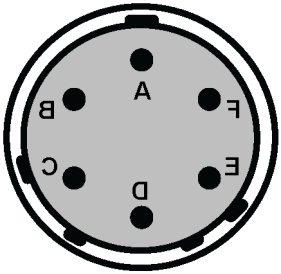


Figure 4 WAA151 connector cable side

1. Connect the heating element in the shaft tunnel between pins D and E.
2. Supply the heating element with 20 V DC or V AC.

3.4 Verifying installation

1. Make sure that the sensor is powered up and connected to the data collection system.
2. Rotate the cup wheel manually and make sure the speed readings change.

4. Maintenance

4.1 Cleaning the anemometer

Heavy contamination in the cups, such as bird droppings or ice, deteriorates the anemometer accuracy. Clean the cups when contaminated.

4.2 Testing anemometer operation

The sensor holds its accuracy in all conditions for 1 year. If the rains are mostly casual and moderate, and the atmospheric corrosion is typical, the sensor accuracy remains for 2 years.

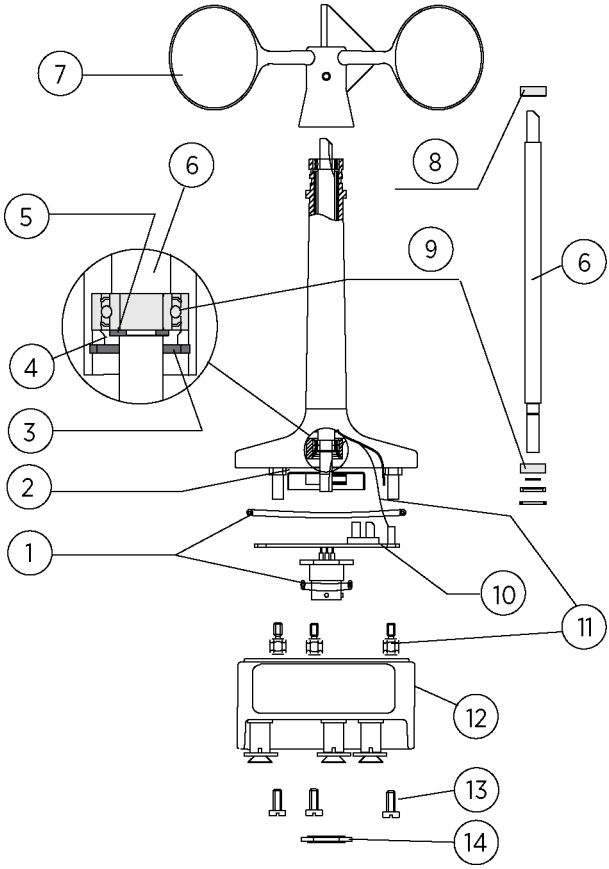
Check the ball bearings and rotate the sensor shaft once a year.

1. Remove the cup wheel.
2. Make sure that the shaft spins smoothly and that there is no noise.

4.3 Replacing consumables



Only a trained technician may replace the bearings.



1 O-ring (2 pcs)
2 Chopper disc
3 External retaining ring, body
4 Spacer ring
5 Internal retaining ring, shaft
6 Shaft and upper body assembly
7 Cup wheel assembly
8 Ball bearings
9 Ball bearings
10 Printed circuit board (PCB)
11 Spacer (3 pcs)
12 Lower body
13 M6x16 DIN7991 (3 pcs)
14 Hex nut of the connector

1. Open the cup wheel fixing screw with a 2-mm Allen key. Remove the cup wheel assembly.

> **CAUTION!** Do not remove the fixing screw to ensure perfect sealing after reassembling. The fixing screw has been treated with sealant.

2. Loosen the hex nut of the connector with a 22-mm tool.

> **CAUTION!** Be careful not to bend the connector pins.

3. Loosen the three pan head screws at the bottom of the sensor body with a 7-mm tool.
4. Remove the lower body assembly by pulling it straight outwards.
5. Loosen the spacer screws with a 7-mm tool and disconnect the heating element outlet.
6. Remove the printed circuit board including the opto-coupler.

> **CAUTION!** Do not twist or bend the connector. This may break the pins.

7. Loosen the fixing screw of the chopper disc with a 2-mm Allen key and remove the chopper disc.
8. Remove the external retaining ring with narrow-pointed pliers.
9. Remove the spacer ring.
10. Remove the internal retaining ring at the bottom of the shaft with narrow-pointed pliers.
11. Remove the lower bearing.
12. Push out the shaft downwards through the upper body.
13. Remove the top bearing after pulling out the shaft.

4.4 Reassembling the anemometer

1. Attach the top bearing.
2. Push the shaft upwards through the upper body.
3. Attach the lower bearing.
4. Attach the internal retaining ring at the bottom of the shaft.
5. Attach the spacer ring.

6. Attach the external retaining ring.
7. Attach the chopper disc onto the shaft. Make sure the disc is positioned so that the disc teeth do not touch the opto-coupler on the circuit board. Tighten the screw.

CAUTION! Make sure that the chopper disc teeth do not touch the opto-coupler. There must be 1 ... 2 mm space between the bottom of the opto-coupler and the disc teeth.

Be careful when handling the new ball bearings. Do not drop them or force them onto the shaft.

8. Attach the heating element outlet to the circuit board. Put the circuit board in place and fasten it with spacers.
9. Put the lower body assembly carefully into place.

   Make sure that the bigger O-ring between the upper and lower sensor bodies is in place and fasten the three screws at the bottom of the sensor.

Vaisala recommends replacing the O-ring with a new one when opening the assembly for maintenance.

10. Tighten the hexagon nut of the connector.
11. Connect the cable plug to the sensor body connector. Fasten the sensor body on the cross-arm with three screws.
12. Mount the cup assembly onto the sensor body. Tighten the fixing screw.

CAUTION! Replacing the heating resistance element requires special tools. To avoid any damages, Vaisala recommends that the manufacturer replaces the heating element.

5. Troubleshooting

Table 3 Troubleshooting the Anemometer

Problem	Probable Cause	Remedy
The data is not received from the sensor.	The sensor is mechanically damaged.	Check the cables and connectors.
	After removing the connector's hexagon nut, the connector is bent, which breaks the connection wires.	Check the connector.
	The sensor is not powered properly.	Make sure that the supply voltage is 9.5 ... 15.5 VDC.
	Some Vaisala products such as WAT12 switch power on to the sensor only for a very short period (200 s to 500 s).	Make sure that the sensor output rises above U_{in} - 1.5 V at the end of the power pulse.
The sensor shaft is covered with ice and snow.	The heating element does not function.	Send the sensor to Vaisala for repair. Contact Vaisala Technical Support.
	The heating element is not properly connected.	Open the sensor and make sure that the heating element outlet is connected to the connector on the printed circuit board. See Replacing consumables (page 14).
The output from connector pin C tracked with an oscilloscope is not square wave.	The printed circuit board is damaged.	Replace the printed circuit board. See Replacing consumables (page 14) and Spare parts and accessories (page 21) for spare part order code.
The frequency is not equal to 14 times revolution rate.		
Current consumption is more than 28 mA when the shaft is rotated mechanically.		

6. Technical data

6.1 WAA151 specifications

Table 4 WAA151 measurement performance

Property	Description/Value
Sensor/Transducer type	Cup anemometer/opto-chopper
Observation range	0.4 ... 75 m/s (0.9 ... 168 mph)
Starting threshold [1)]	< 0.5 m/s (1.1 mph)
Distance constant	4.0 m (13 ft 1 in)
Transducer output	
0 ... 75 m/s (0 ... 168 mph)	0 ... 750 Hz square wave
Characteristic transfer function	Uf (wind speed) = 0.328 + 0.101 × R (output pulse rate)
Transducer output level	
(I_{out} < +5 mA)	High state > U_{in} –1.5 V
(I_{out} > -5 mA)	Low state < 2.0 V
Accuracy within 0.4 ... 60 m/s (0.9 ... 134 mph)	
With characteristic transfer function (standard deviation)	±0.17 m/s (0.38 mph)
With simple transfer function U_f = 0.1 × R	±0.5 m/s (1.12 mph)

1) *Measured with the cup wheel in position least favored by flow direction. The optimum position yields a < 0.35 m/s (0.8 mph) starting threshold.*

RANGE (m/s)	0-3	3-10	10-17	17-24	24-31	31-37	37-44	44-51	51-58	58-65
ERROR (m/s)	-0.4	-0.3	-0.2	-0.1	0.0	+0.1	+0.2	+0.3	+0.4	+0.5

Table 5 WAA151 inputs and outputs

Property	Description/Value
Electrical connections	MIL-C-26482 type, 6-wire cable
Cabling	6-wire cable through cross-arm
Recommended connector at cable end	Souriau UTS6JC10E6P
Operating power supply	U_{in} = 9.5 ... 15.5 V DC, 20 mA, typical

Property	Description/Value
Heating power supply	AC or DC 20 V, 500 mA, nominal
Settling time after power-up	< 30 µs

Table 6 WAA151 operating environment

Property	Description/Value
Operating temperature [1)]	−50 ... +55 °C (−58 ... +131 °F)
Storage temperature	−60 ... +70 °C (−76 ... +158 °F)
Operating humidity	0 ... 100 %RH
IP rating	IP65: Dust-tight. Protected from water jets from any direction.

1) With shaft heating.

Table 7 WAA151 mechanical specifications

Property	Description/Value
Dimensions (H × Ø)	240 × 90 mm (9.45 × 3.54 in)
Swept radius of cup wheel	91 mm (3.58 in)
Weight	570 g (1.26 lb)
Materials	
Housing	AlMgSi, gray anodized
Cup	PA, reinforced with carbon fiber

Table 8 WA15 compliance

Property	Description
Compliance marks	CE, China RoHS
EU directives and regulations	RoHS Directive (2011/65/EU) amended by 2015/863 EMC Directive (2014/30/EU)
Electromagnetic compatibility (EMC)	EM 55032:2015, Electromagnetic compatibility of multimedia equipment - Emission requirements. Class B. EN 61326-1:2013, Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 1: General requirements.

Property	Description
Environmental	IEC/EN 63000:2018

6.2 Spare parts and accessories

Table 9 WAA151 spare parts and accessories

Item	Order code
Crossarm and termination box	WAC151
Crossarm and serial RS-485 transmitter	WAC155
Crossarm and analog transmitter	WAT12
Component board for WAC155	WAC155CB
Component board for WAT12	16637WA
Service kit for one WA15/25 sensor (set of bearings and gasket)	16644WA
6-lead heating power cable (10 m) for WA15/25, open leads on both ends	ZZ45049
16-lead signal cable (10 m) for WA15/25, open leads on both ends	ZZ45048
Special length 16-lead signal cable for WA15/25, open leads on both ends	ZZ45048SPEC
Special length 6-lead heating power cable for WA15/25, open leads on both ends	ZZ45049SPEC
Sensor cable for WAA151/252 0.8 m (31.5 in), open lead on one end (6 wires), connector 230118 on other end	ZZ45036
Sensor cable for WAV151/252 0.8 m (31.5 in), open lead on one end (10 wires), connector 230119 on other end	ZZ45037
Special length sensor cable for WAA151/252, open lead on one end (6 wires), connector 230118 on other end	ZZ45036SPEC
Special length sensor cable for WAV151/252, open lead in one side (10 wires) and connector 230119 in another side	ZZ45037SPEC
Sensor board for WAA151	1433WA
Cup assembly for WAA151	7150WA
Connector for WAA151, WAA252	230118
Power supply for WA15	WHP151
Power board for WHP151 power supply	WA35120
Attachment hardware for WAA151/252 and WAV151/252	16546WA

6.3 Mean time between failures (MTBF)

The following table lists the calculated mean times between failures according to the standard MIL-HDBK-217F. The last column provides the failure frequency during 10^6 hours of use.

Table 10 MTBF values

Abbreviation	Installation	f/10^6 h
Gf	Permanent installations	2.34
Gm	Mobile installations	8.52
Nu	Ship equipment	5.48

6.4 Mean time to repair (MTTR)

Mean time to repair is 0.3 h.

Technical support



Contact Vaisala technical support at helpdesk@vaisala.com. Provide at least the following supporting information as applicable:

- Product name, model, and serial number
- Software/Firmware version
- Name and location of the installation site
- Name and contact information of a technical person who can provide further information on the problem

For more information, see www.vaisala.com/support.

Warranty and product returns

For standard warranty terms and conditions, see www.vaisala.com/warranty.
Please observe that any such warranty may not be valid in case of damage due to normal wear and tear, exceptional operating conditions, negligent handling or installation, or unauthorized modifications. Please see the applicable supply contract or Conditions of Sale for details of the warranty for each product.

1. Read the warranty information.
2. Contact Vaisala technical support and request a Return Material Authorization (RMA) and shipping instructions.



Always request the RMA before returning any faulty material.
Provide the failure report as requested.

Recycling



Recycle all applicable material according to local regulations.

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